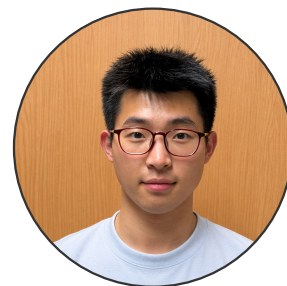


Kaicheng Mao

DATA SCIENCE STUDENT · AI RESEARCH

Shenzhen, China

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Education

SUSTech(Southern University of Science and Technology)

Shenzhen, China

B.S. IN DATA SCIENCE

Sept. 2023 - June. 2027

- GPA: 3.74/4.0 (89.98/100)
- Relevant Coursework: Advanced Natural Language Processing, Data Structures and Algorithm Analysis, Distributed Storage and Parallel Computing, Statistical Linear Model, Data Science Practice(Deep Learning)

Shenzhen Middle School

Shenzhen, China

HIGH SCHOOL DIPLOMA

Sept. 2020 - June. 2023

Skills

Programming Python, Java, R

Machine Learning LLM post-training

English IELTS 7.5 (Listening 8.5, Reading 9.0, Writing 6.5, Speaking 6.5), CET-4: 632, CET-6: 584

Sports Volleyball, Weightlifting, Powerlifting, CrossFit

Project Experience

Action Memory: Structured Memory for Scientific AI Agent Citation Consistency

SCIENTIFIC AGENT EVALUATION RESEARCH

Apr. 2026 - Jun. 2026

- Designed and implemented JSONL-based Action Memory that converts historical citation error repair experience into structured procedural caution, supporting BM25/TF-IDF/Hybrid retrieval with bilingual tokenizer and Top-6 memory injection before generation.
- Action Memory reduced citation error rate from 28.2% to 5.1% (↓82%) with 3 fully traceable causal chains (audit write → retrieval hit → claim correction); structured Memory reduced total errors from 22 to 16 (↓27%), eliminating Contradictions in main experiment.

Deep Learning for Precipitation Nowcasting

DEEP LEARNING RESEARCH PROJECT

Feb. 2026 - Present

- Enhanced the SEVIR dataset with NASADEM-derived elevation, slope, aspect, and land-sea information.
- Built geospatial preprocessing foundations for future multi-source data fusion in precipitation nowcasting models.
- Prepared terrain-aware auxiliary inputs to support downstream deep learning experiments for short-term precipitation forecasting.
- Developed a Cross-Attention and FiLM-based UNet that fuses elevation, radar, and infrared multi-source data for more comprehensive precipitation nowcasting.

Fine-tuning Qwen3-0.6B-Base on Nemotron-CC-Math

LLM FINE-TUNING PROJECT

Mar. 2026 - May. 2026

- Fine-tuned Qwen3-0.6B-Base on Nemotron-CC-Math for mathematical domain adaptation.
- Designed a data cleaning pipeline using only regular expressions and Qwen3 itself, without teacher models, tool-calling, search, or external tools.
- Focused on lightweight, self-contained data filtering and refinement to support reproducible LLM post-training experiments.
- Designed reward functions targeting Weird Token generation and Critical Thinking deficiency, then applied GRPO to improve accuracy from a 40.2% baseline to 56.1%, validating the core idea of the 2025 NeurIPS paper "Does Reinforcement Learning Really Incentivize Reasoning Capacity in LLMs Beyond the Base Model?"

Intelligent Scissors Project

IMAGE SEGMENTATION PROJECT

Apr. 2025 - May. 2025

- Implemented an interactive image segmentation system combining gradient-based edge detection with dynamic pathfinding.
- Applied Sobel operators to compute directional gradients and construct a weighted graph, assigning lower edge costs to stronger image boundaries.
- Used Dijkstra's algorithm for real-time optimal path calculation between user-selected points, with adaptive path stabilization and cursor snapping to maximum-gradient edges within a 5-pixel radius.

Extracurricular Activity

GEM Trailblazer Summer, Nanyang Technological University

SUMMER PROGRAM PARTICIPANT

Singapore
Jun. 2024 - Jul. 2024

- Completed coursework in Cities and Urban Life and Introduction to Data Science & Artificial Intelligence.
- Collaborated and built connections with peers from diverse educational and cultural backgrounds.

SUSTech Volleyball Team

STARTING SETTER

Shenzhen, China
May 2024 - Now

- Served as the starting setter for the Southern University of Science and Technology varsity volleyball team.
- Competed in the 14th Guangdong University Volleyball League, the 15th Guangdong Volleyball Championship, and the Shenzhen Intercollegiate Sports Competition volleyball event (third place).

SUSTech Volleyball Club & Varsity Volleyball Team

PRESIDENT & UNDERGRADUATE CAPTAIN

Shenzhen, China
Sept. 2025 - Aug. 2026

- Led the SUSTech Volleyball Club as president and served as undergraduate captain of the varsity volleyball team.

Honors & Awards

2025	Most Valuable Player , The 7th SUSTech Volleyball Championship (Student Division)	Southern University of Science and Technology
2025	Outstanding Student , Exceptional performance in 2025	Southern University of Science and Technology
2024-2025	Third-class Merit Student Scholarship , Southern University of Science and Technology	Shenzhen, China
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